

CHG 569

Environmental Pollution Engineering

Module V-Z

(Managing Air Pollution-CONTD)



Methods of air pollution Control

► Treatment (add-on controls)

Four commonly used control technologies for gaseous pollutants:

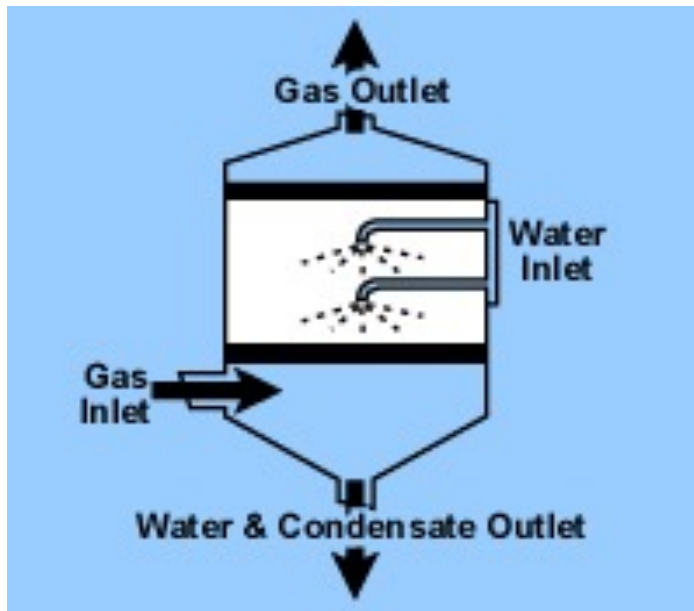
- Absorption ☒
- Adsorption ☒
- Condensation, and
- Incineration (Combustion).

Condensation

- ▶ Temperature of gas stream is reduced to $<$ saturation temperature of gaseous pollutant to be removed by the use of a cold medium (cold wall of a heat exchanger or fluid)
- ▶ Condensers are simple, relatively inexpensive devices that normally use water or air to cool and condense the pollutant to be removed.
- ▶ used as ***pretreatment devices***, ahead of adsorbers, absorbers, and incinerators to reduce the total gas volume to be treated by more expensive control equipment. Also to remove corrosive or damaging gases.

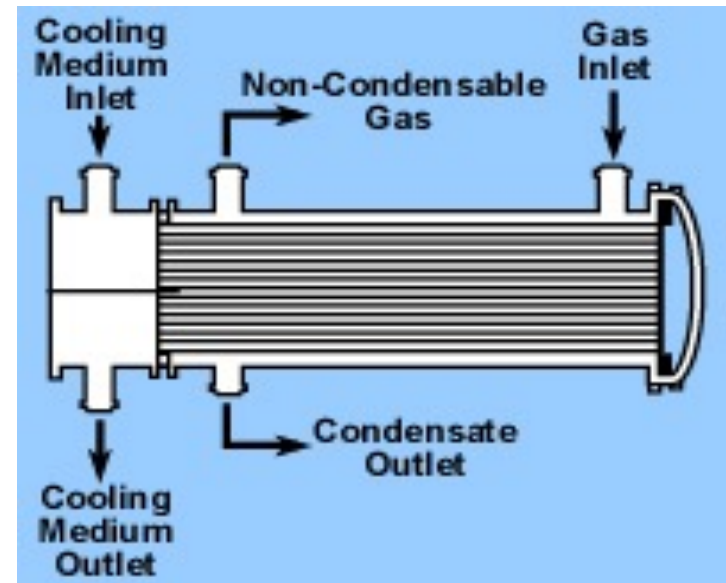
Two types of condensers

contact condensers



the gas comes into contact with cold liquid

and surface condensers



the gas contacts a cooled surface in which cooling medium is circulated tube

Removal efficiencies typically range from 50 – 95+% depending on design and applications.

Design consideration for Condenser

- Surface area for surface condensers

The discharge gas stream from a meat processing plant consist mainly of atmospheric steam, plus a small fraction of noncondensable odourous gases. The stream is to pass through a condenser to remove the steam before the noncondensable gases go to either an incinerator or an adsorber.

Estimate the size of a condenser to treat 10kg/s of this discharge stream.

Information Given

Overall heat transfer coefficient	– 765W/(m ²)(°C)
Enthalpy of vapourisation (Steam)	- 2326kJ/kg
Cooling water	-enters at 27°C and leaves at 46°C

Design consideration for Condenser- contd

Sizing procedure

1. Determine the heat load Q for the condenser

$$Q = (\text{Enthalpy of vapourisation}) \times (\text{flowrate})$$

$$= 2326 \times 10 \frac{\text{kJ}}{\text{kg}} \cdot \frac{\text{kg}}{\text{s}} = 23.26 \text{ kW} = 23.26 \times 10^6 \text{ W}$$

2. Estimate the log-mean temperature difference driving force

$$\text{LMTD, } \Delta T_{LM} = \frac{\Delta T_L - \Delta T_O}{\ln \frac{\Delta T_L}{\Delta T_O}} = \frac{(100 - 27) - (100 - 46)}{\ln \frac{(100 - 27)}{(100 - 46)}} = \frac{73 - 54}{\ln \frac{73}{54}} = \frac{19}{0.30} = 63^\circ \text{C}$$

3. Calculate the required area of the condenser

$$Q = UA\Delta T_{LM} \Rightarrow A = \frac{Q}{U\Delta T_{LM}} = \frac{23.26 \times 10^6 \text{ W}}{765(63)} \frac{\text{m}^2 \cdot ^\circ \text{C}}{\text{W} \cdot ^\circ \text{C}} = 482.6 \text{ m}^2$$

Combustion (Incineration)

- To control the emissions of combustible organic compounds from process industries

complete combustion → gaseous stream is converted to carbon dioxide and water vapor

Incomplete combustion → pollutants being released into the atmosphere and smoke

*Harmful oxidable gases → Harmless products

- Used for
 - highly odourous gases (e.g. mercaptans, cyanide gases and hydrogen sulphide)
 - Organic aerosols (which cause visible plumes)
 - Highly combustible waste gases produced by industries
- Combustion = Fuel + oxygen + source of ignition
- Combustion takes place in equipment known as afterburners or incinerators or combustors

3 methods of combustion

- Direct combustion or flaring,
- Thermal incineration, and
- Catalytic incineration.

Which method to choose depends on some factors:

- ▶ type of pollutants in the waste stream,
- ▶ concentration of combustibles in the stream,
- ▶ Inlet stream temperature
- ▶ inlet stream volumetric flow rate
- ▶ permissible emission levels for the pollutant, and an
- ▶ economic evaluation.

Combustion - contd

In combustion equipment, oxidation reactions are pushed as close as possible to completion leaving a minimum of unburnt compounds.

Efficiency of afterburners

Based on HCs alone
$$\frac{HC_{in} - HC_{out}}{HC_{in}} \cdot 100$$

Based on Los Angeles County Rule 66

$$\frac{HC_{in} - [HC_{out} + (CO_{out} - CO_{in})]}{HC_{in}} \cdot 100$$

To achieve complete combustion, the afterburner must provide the proper amount of turbulence and burning time, and it must maintain a sufficiently high temperature.

The normal range for the 3Ts are:

Temperature:	375 - 825 °C
Residence time:	0.2 - 0.6 sec
Gas velocity:	4 - 8 m/sec

Direct Combustion or Flaring

A direct combustor or flare is an open ended combustion unit, maintained at the end of a stack discharging the gas stream

- Gas stream must have high energy concentration so that it is self-supporting (no need for additional fuel) after ignition.

$$\text{LFL} < \text{conc. of pollutant} < \text{UFL}$$

- The least costly for organic pollutants
- Occasional smoky or sooty appearance

The black smoke, which is mainly made up of carbon particles, occurs as a result of incomplete combustion when there is insufficient amount of air to support the burning of gases sent to the flare system.

Steam is usually injected to the flame zone to increase turbulence in the burning region. This increases the air intake which is necessary for complete combustion and provides better contact of oxygen with the combustible pollutant .

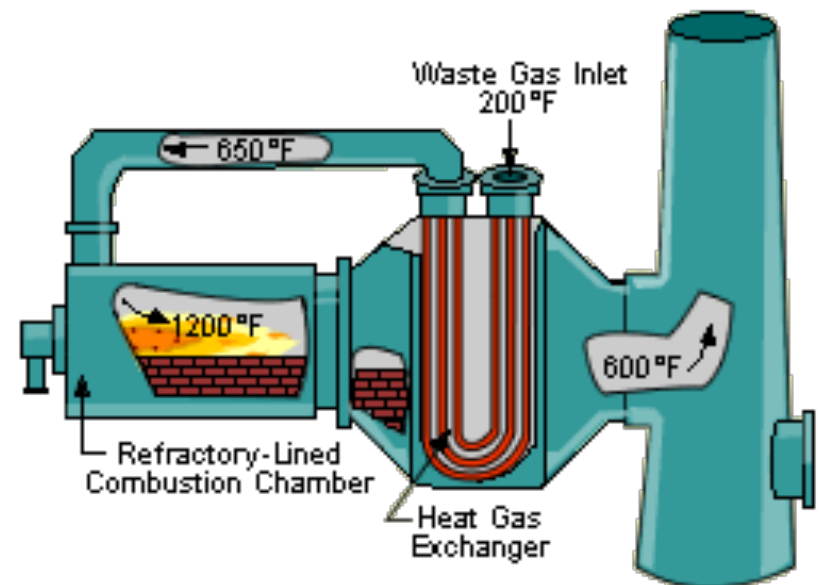
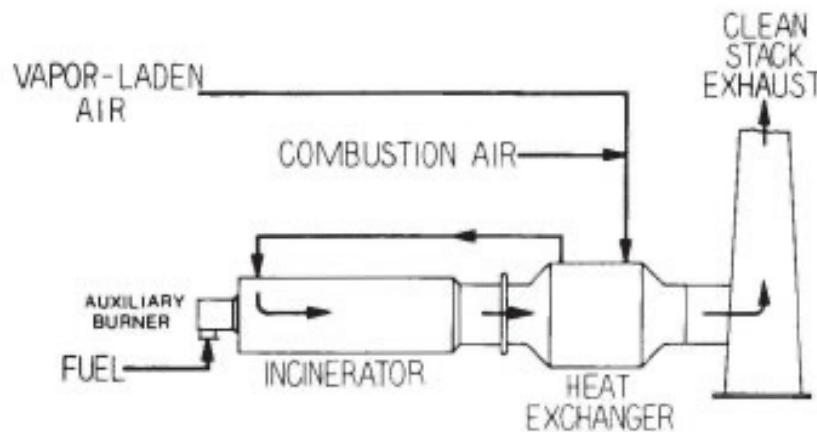
Result is smokeless flaring.



Thermal incineration

- Used when the conc. of combustible pollutant is low ($< \text{LFL}$), therefore additional fuel is required.

The waste gas stream is preheated in a heat exchanger and then passed thru the combustion zone of a burner supplied with supplemental fuel



Thermal incineration - contd

Important considerations in the design of thermal incinerators

The 3Ts of combustion

1. **Residence Time** sufficient to allow complete combustion. Normally in the range 0.2 – 0.8sec (0.5secs reasonable)

2. **Temperature** a function of the combustible pollutant in the gas stream

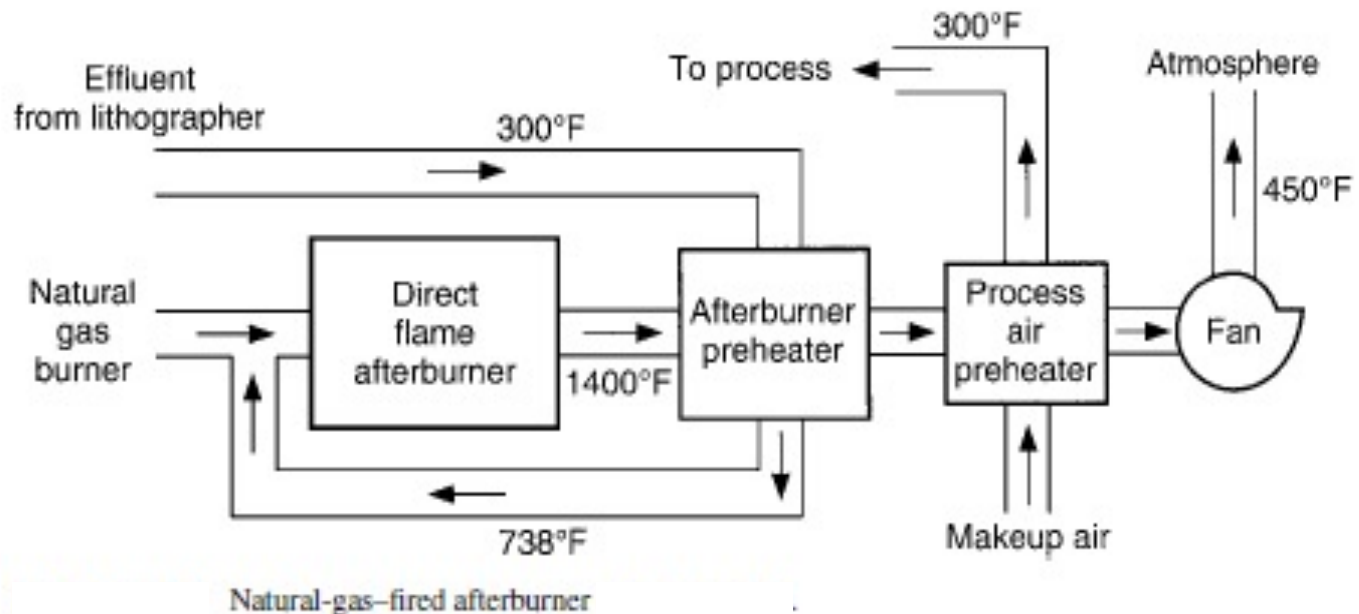
Process	Average temp range (K)
HC oxidation	780 - 1030
CO oxidation	950 - 1060
Odour control via oxidation	900 - 1300

3. **Turbulence** amount of mechanical mixing to ensure complete contact of oxygen with fuel (combustible pollutant). If there is proper mixing then less residence time is required.

and of course **oxygen** (assumed in turbulence)

PERFORMANCE OF AN AFTERBURNER

It is proposed to use a natural-gas-fired, direct-flame afterburner to incinerate toluene in the effluent gases from a lithography plant. The afterburner system is as shown below. The flow rate of the 300°F (422 K) effluent is 7000 std ft³/min (3.30 m³/s), standard conditions being 60°F and 1 atm; its toluene content is 30 lb/h (0.0038 kg/s). After passing through the afterburner preheater, the gas enters the afterburner at 738°F (665 K). The afterburner is essentially a horizontal cylinder, 4.2 ft (1.28 m) in diameter and 14 ft (4.27 m) long; it can be assumed to incur heat losses at 10 percent in excess of the calculated heat load. Gases leaving the afterburner are at 1400°F (1033 K).



When reviewing plans for such an installation, the local air-pollution-control agency knows from experience that in order to meet emission standards, the afterburner must operate at 1300 to 1500°F, that the residence time in the vessel must be 0.3 to 0.5 s, and that the velocity within it must be 20 to 40 ft/s. Can this afterburner satisfy those three criteria?

Use the following data:

Gross heating value of natural gas: 1059 Btu/std ft³

Volume of combustion products produced per standard cubic foot of natural gas burned: 11.5 std ft³ (11.5 m³ per standard cubic meter)

Average available heat from natural gas between 738 and 1400°F: 600 Btu/std ft³ (22,400 kJ/m³)

Molecular weight of toluene: 92

Average heat capacity C_{p1} of effluent gases between 0 and 738°F: 7.12 Btu/(lb · mol)(°F)[29.8 kJ/(kg · mol)(K)]

Average heat capacity C_{p2} of effluent gases between 0 and 1400°F: 7.38 Btu/(lb · mol)(°F)[30.9 kJ/(kg · mol)(K)]

Volume of air required to combust natural gas: 10.33 std ft³ air/std ft³ natural gas (10.33 m³/m³ natural gas)

Calculation Procedure

1. Convert the gas flow rate to the molar basis. Since 1 lb-mol of gas at 32°F and 1 atm occupies 359 ft³, at 60°F it will occupy $(359)[(460 + 60)/(460 + 32)]$, i.e., 379 ft³. Accordingly, the molar gas flow rate $\dot{n}$ is $(7000 \text{ actual ft}^3/\text{min})/(379 \text{ actual ft}^3/\text{lb} \cdot \text{mol})$, i.e., 18.47 lb · mol/min (0.139 kg · mol/s).

2. Calculate the total heat load (heating rate) required to raise the gas stream from 738 to 1400°F. Since the heat capacity data are given with 0°F as a basis, this heat load Q must be based on first cooling the gases from 738 to 0°F, then heating them from 0 to 1400°F. Thus,

$$\begin{aligned} Q &= \dot{n}[C_{p2}(1400 - 0) - C_{p1}(738 - 0)] \\ &= 18.47[(7.38)(1400) - (7.12)(738)] = 93,790 \text{ Btu/min (1648 kW)} \end{aligned}$$

3. Determine the actual required heat load, taking into account the 10 percent heat loss. Thus, actual heat load = $1.1Q = (1.1)(93,790) = 103,169 \text{ Btu/min (1813 kW)}$.

4. Find the rate of natural gas needed to satisfy this heat load. This is, simply, the heat load divided by the available heat. Thus,

$$\text{Rate of natural gas} = 103,169/600 = 171.9 \text{ std ft}^3/\text{min (0.081 m}^3/\text{s)}$$

5. Determine the volumetric flow rate of the combustion products of the natural gas. This flow rate q_1 equals the natural gas rate times the volume of combustion products produced per standard cubic foot of natural gas. Thus,

$$q_1 = (171.9)(11.5) = 1976 \text{ std ft}^3/\text{min (0.932 m}^3/\text{s)}$$

Note that the 11.5 figure already takes into account the air needed to burn the natural gas.

6. Calculate the total volumetric flow rate through the afterburner. This flow rate q_T equals that of the effluent from the lithography plant plus that of the combustion products from step 5. Thus,

$$q_T = 7000 + 1976 = 8976 \text{ std ft}^3/\text{min} (4.233 \text{ m}^3/\text{s})$$

For the subsequent steps, this figure must be converted to actual cubic feet per minute. Since the afterburner operates at 1400°F, this equals $(8976)[(1400 + 460)/(60 + 460)]$, or 32,106 actual ft³/min (15.14 m³/s).

7. Determine the cross-sectional area of the afterburner. As the cross section is circular, this area S equals $\pi D^2/4$, where D is the diameter of the afterburner. Thus, $S = \pi(4.2)^2/4 = 13.85 \text{ ft}^2$ (4.22 m²).

8. Calculate the velocity through the afterburner. This residence time t equals the length of the vessel divided by the velocity. Thus, $t = 14/38.6 = 0.363 \text{ s}$.

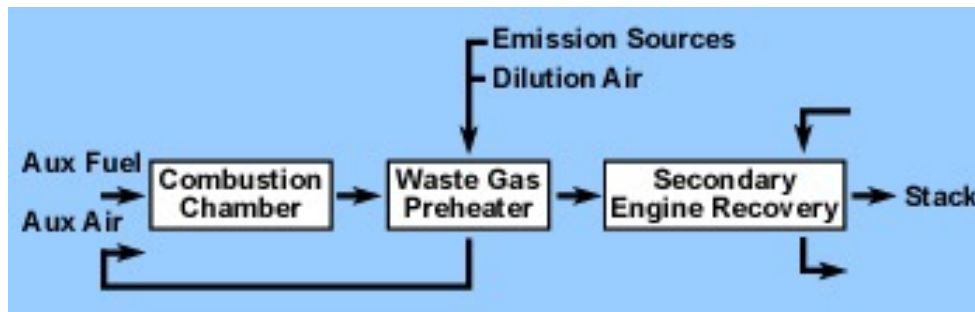
9. Find the residence time within the afterburner. This residence time t equals the length of the vessel divided by the velocity. Thus, $t = 14/38.6 = 0.363 \text{ s}$.

10. Ascertain whether the afterburner meets the agency's criteria. The afterburner operates at 1400°F (1033 K), so it meets the temperature criterion. The residence time of 0.363 s meets the residence-time criterion. And the gases pass through the vessel at 38.6 ft/s (11.8 m/s), which satisfies the velocity criterion. Thus, all three criteria are satisfied.

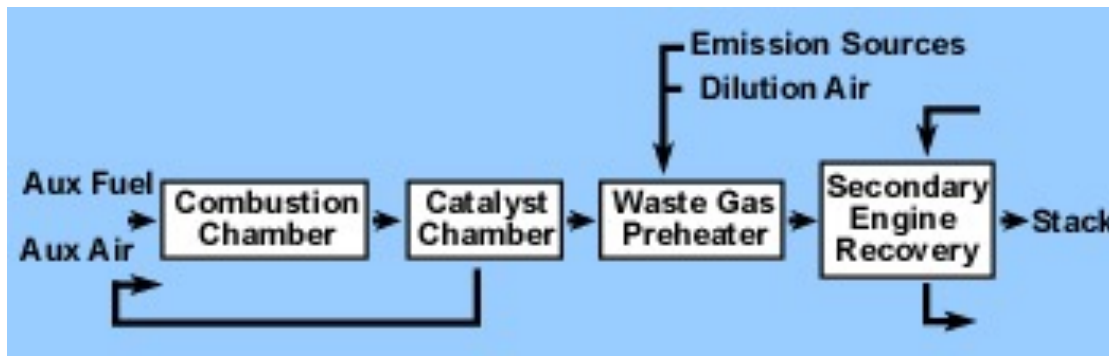
Catalytic incinerators

- Catalytic incinerators are similar to thermal incinerators.

The main difference is that after passing through the flame area (see next slide), the gases pass over a catalyst bed. This promotes combustion at lower temperatures, thereby reducing fuel costs, and lowers the residence time.



← Thermal incinerator



← Catalytic incinerator

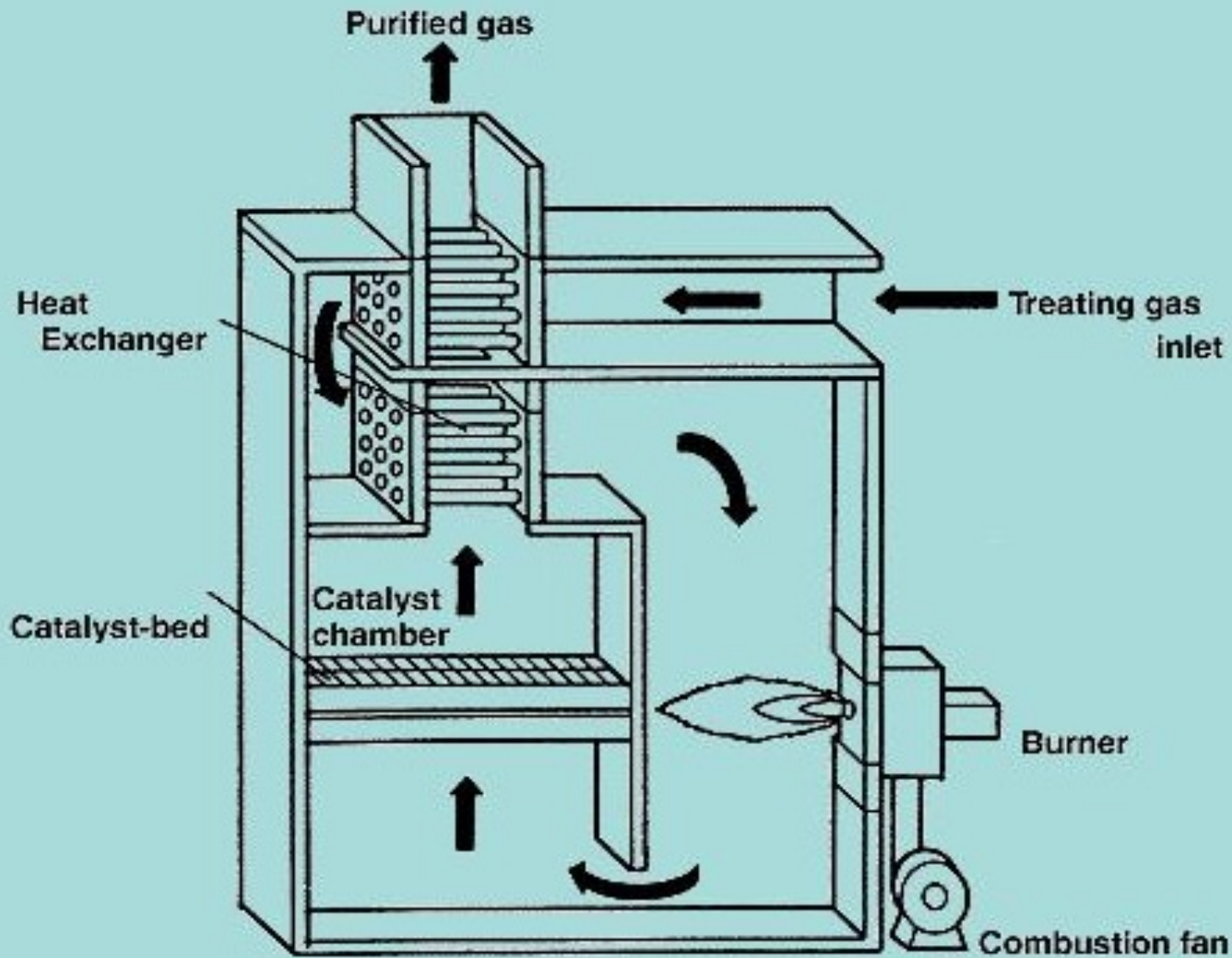


Diagram of a typical catalytic incinerator

Catalytic incineration – contd.

- Residence time <<< than that for thermal units (~0.02-0.09secs).

Therefore instead of residence time **space velocity** is used $\frac{\text{ft}^3/\text{hr of gas}}{\text{ft}^3 (\text{catalyst volume})}$

- catalysts are platinum and palladium.
- Ignition temperature is lowered (315 – 485°C)
- Oxidation (Combustion) reaction occurs on the catalyst surface

Combined Heat and Power (CHP) or Cogeneration

In most combustion plants (either for electricity generation or otherwise.) a certain amount of heat is released (wasted) into the environment (up to two thirds of the overall energy consumed is lost) through cooling towers (often seen as a cloud of steam rising from the towers), flue gas, or by other means.

CHP captures some of this heat (cutting waste) and converts it to **heat** (hot water) and **power** (electricity) simultaneously.

Thereby increasing the overall efficiency of the plant to about 80%

Fuel oil
Storage.

Waste gas to be
Combusted

Air

Combustion
chamber

Flue gas

Compressed

Power
turbine

Alternator

Electricity
to site.

Feedwater

Waste
heat
boiler

Steam to site.

Stack.

Catalytic Converter
 $\text{NO}_x = \text{N}_2 + \text{O}_2$
 $\text{CO} = \text{CO}_2$

Abatement Control technologies for NO_x control

Preventive

- Flue gas recirculation ☒
- Staged combustion ☒

Add-on Controls

- ❖ **Selective Catalytic Reduction (SCR)** - uses a catalyst bed (vanadium-titanium or platinum based and zeolite and ammonia (NH₃) or Urea



- ❖ **Selective Non Catalytic Reduction (SNCR)** - ammonia or urea is injected into the flue gas at a temperature 870 – 1090°C. Urea is converted to ammonia, which reacts to reduce NO_x to N₂ and H₂O
- ❖ **Non Selective Catalytic Reduction (N SCR)** - uses a three way catalytic converter, where NO_x, HCs, CO → N₂, O₂ and H₂O

Biofilters – use of microorganisms

Biofiltration is the biodegradation of gaseous pollutants by aerobic microorganisms attached, as a thin layer of moisture (BIOFILM), to a medium. Mainly used for malodourous gases and low concentrations of VOCs. The microorganisms are used to completely oxidize organic compounds into CO_2 and H_2O .

Organic & inorganic pollutant + O_2 -----→ CO_2 + H_2O + heat + biomass

Components of a Biofilter

- **Substrate material** used as support/bed media for microorganisms to attach themselves and provide elemental nutrients for the microbes e.g compost, bark, coarse soil. Synthetic support media are ceramics and plastics.
- **Support rack** – perforated to allow air from below move to the bed media to contact the microbes that live in the bed, also allows excess moisture to drain out
- **Fan** – used to collect the polluted air stream

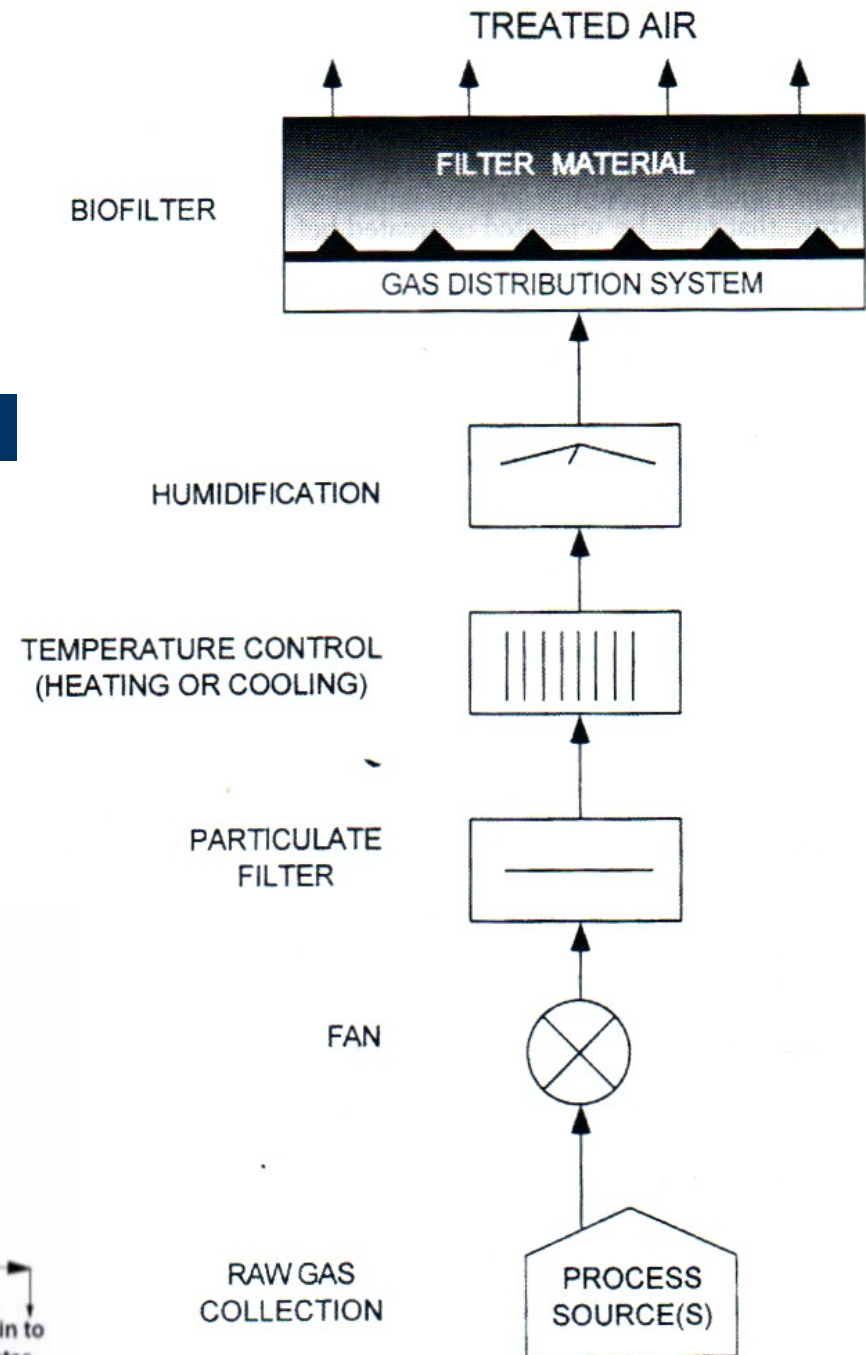
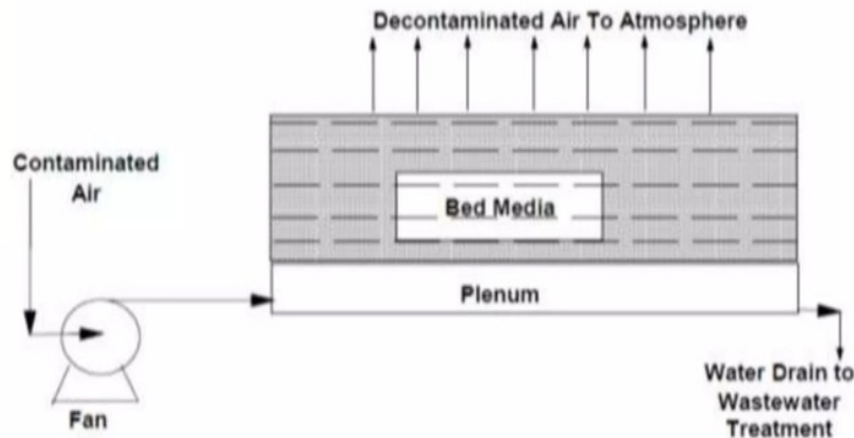
Factors to be considered in a biofilter

- **Polluted gas stream composition**
 - Pollutant concentration below lethal levels for the microorganisms used
 - Oxygen concentration equal to ambient level
- **Moisture content** – Microorganisms need a moist environment, optimum is 20 – 60%
- **Temperature** - Microorganisms operate best between 20°C and 40°C
- **pH** - Microorganisms are most efficient at neutral pH
- **Nutrient supply** – for aerobic microorganisms, the O:N:P ratio is estimated at 100:5:1 (oxygen:nitrogen:phosphorus) and some trace minerals
- Preconditioning to **remove particulates** tht can damage pore structure of filter bed and increase pressure drop

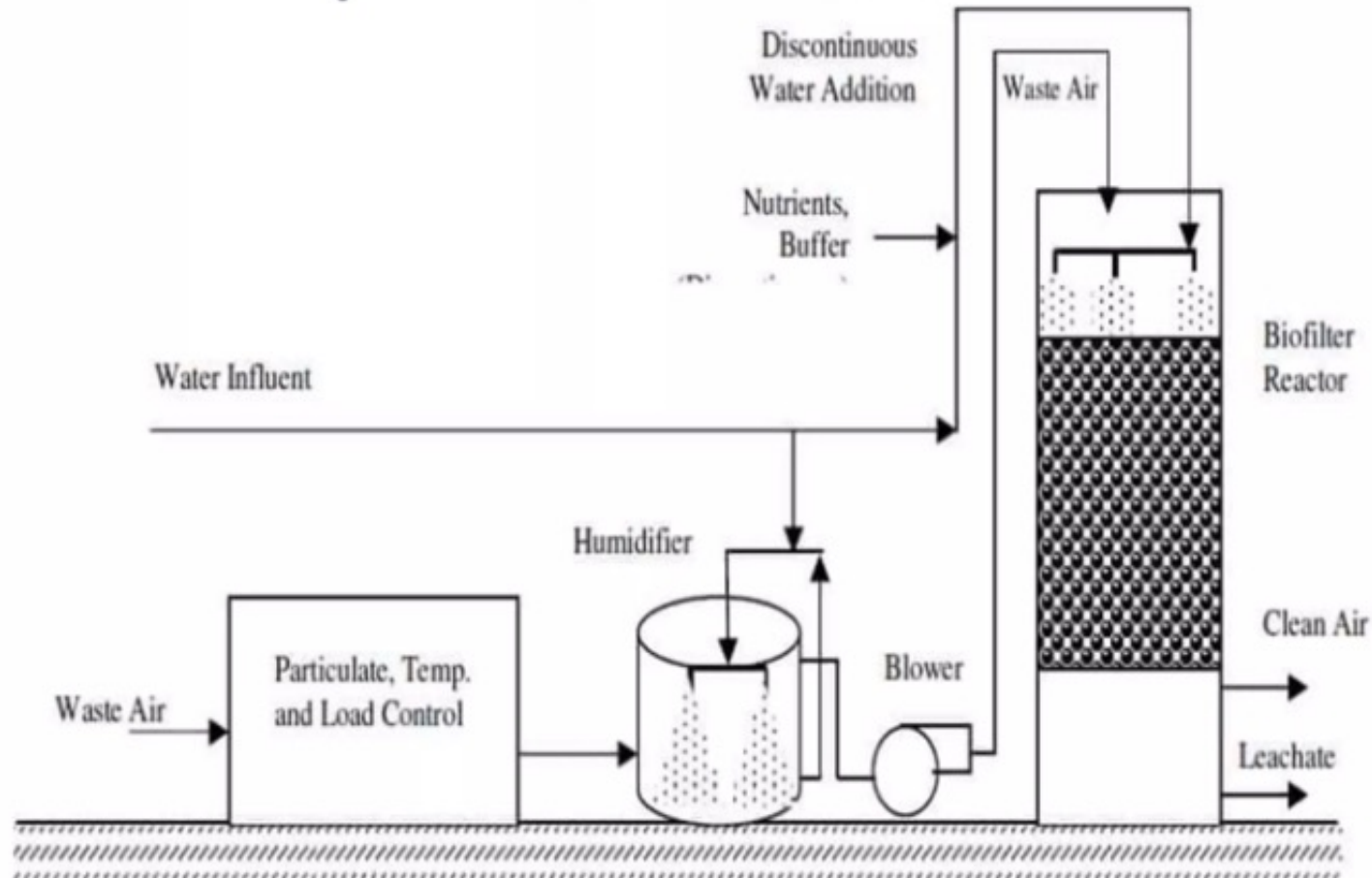
Biofilters are most economical for low concentrations (<1000 ppm) that are oxygen rich. And up to 90% efficiencies are achievable.

Schematic Diagram of a Biofilter Unit

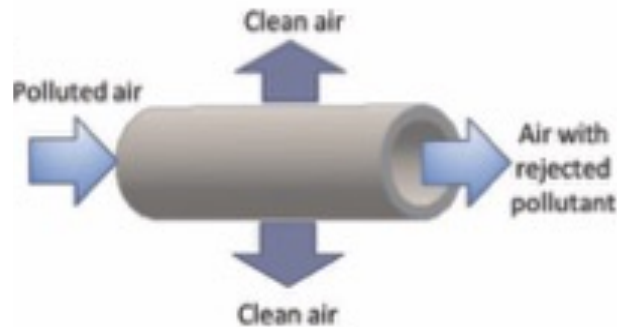
As the polluted gas stream flows through the porous bed media, the pollutants are absorbed by moisture on the bed media (diffuse into the biofilm) where they get decomposed by natural aerobic decomposition process.



Components of Biofilter Unit



Membrane filtration — use of selectively permeable membrane, fabricated from polymeric materials,



Membrane filtration has been used successfully for

- VOC removal from air
- Acid gas removal from flue gas
- NO_x removal from fuel combustion

What are the advantages and disadvantages of the air pollution control technologies, discussed in class, for gaseous pollutants

Reference: **Perry's Chemical Engineers' Handbook**

Chapter 25: Air-pollution management of stationary sources

Methods of air pollution Control

▶ Treatment (add-on controls)

- Gaseous pollutants ☒

- Particulate pollutants

The choice of control device depends on

☛ the characteristics of the particulate exhaust stream, and includes:

- ▶ range of particle sizes ,
- ▶ the exhaust flow rate, temperature, and moisture content.
- ▶ chemical properties such as flammability, acidity, alkalinity, etc.
- ▶ specific characteristics of the site.

☛ The collection efficiency required

The conventional devices used for controlling particulate emissions are:

- Wet collectors - Venturi scrubbers
- Centrifugal Separators - Cyclones
- Gravity settling chambers
- Electrostatic precipitators, and
- Fabric filters.

Collection Mechanism

- "tools" that can be used for separating particles from the gas stream

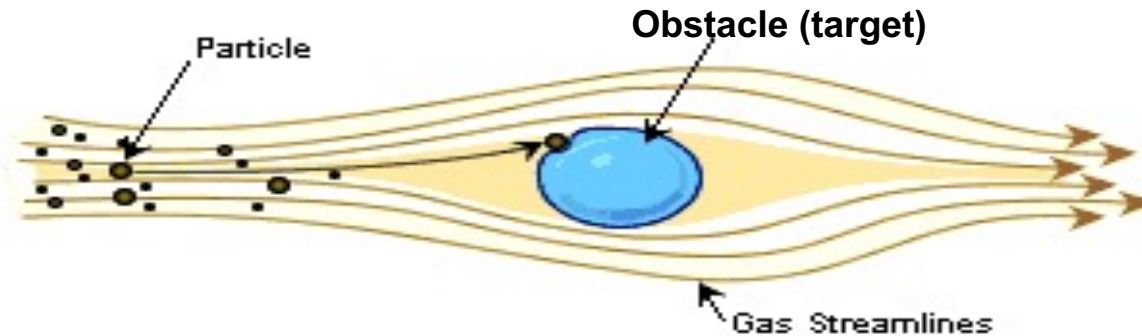
- Inertial impaction and interception
- Brownian diffusion
- Gravitational settling
- Electrostatic attraction
- Thermophoresis
- Diffusiophoresis

All of these collection mechanism forces are strongly dependent on particle size

One or more of these mechanisms are used in anyone of the control devices mentioned in slide 30

Inertial impaction

Due to inertia, a particle moving in a gas stream can strike slowly moving or stationary obstacles (targets) in its path. As the gas stream deflects around the obstacle, the particle continues toward the object and impacts it



Two primary factors affect the probability of an impaction occurring:

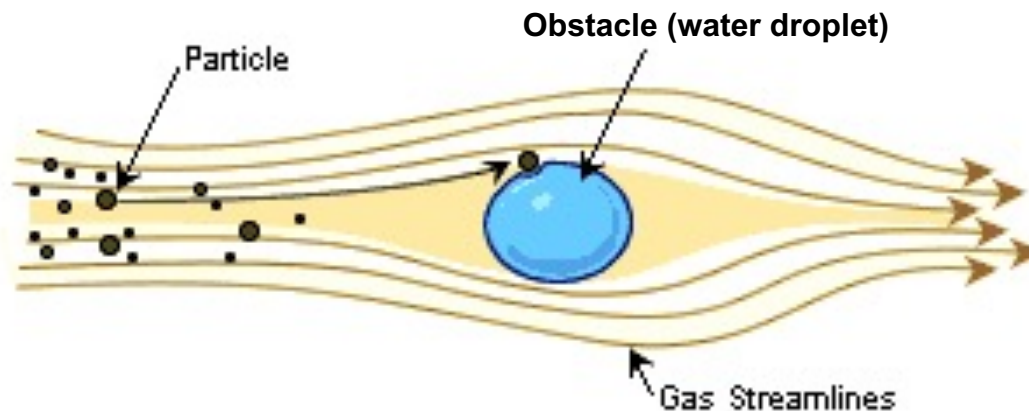
- particle size- as $D_p \uparrow$, collection efficiency $\uparrow$ due to their greater inertia
- the difference in velocity between the particle and the obstacle. $-\uparrow \leftrightarrow \uparrow$

$$\text{Impaction parameter, } k_p = \left[\frac{D_p^2 \rho_p (v_p - v_c) C_s}{18 \mu_g D_c} \right]$$

DIRECT INTERCEPTION

For Inertial impaction, obstacles (e.g. water droplets) are *directly* in the path of the particle moving in the gas stream. Sometimes the obstacle or target is offset slightly from the direct path of the moving particle.

The particle approaches the edge of the obstacle and is collected by **direct interception**

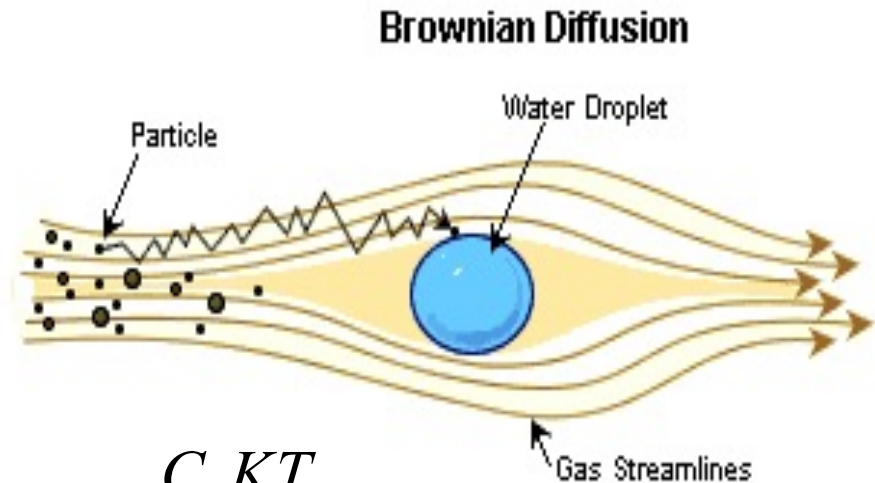


Mechanism of Inertial impaction and interception are highly efficient for $D_p > 10\mu\text{m}$.

BROWNIAN DIFFUSION

Very small particles in a gas stream deflect slightly when gas molecules strike them. Transfer of kinetic energy from the rapidly moving gas molecule to the small particle causes this deflection, called **Brownian diffusion**. These small particles are captured when they impact a target (e.g. liquid droplet) as a result of this random movement.

Diffusivity is a measure of the extent to which molecular collisions influence very small particles, causing them to move in a random manner across the direction of gas flow. The diffusion coefficient in the equation given represents the diffusivity of a particle at certain gas stream conditions.



$$D_d = \frac{C_c KT}{3\pi d_p \mu}$$

Where:

- D_p = Diffusion coefficient (cm²/sec)
- C_c = Cunningham slip correction factor (dimensionless)
- K = Boltzmann constant (gm·cm²/sec²·°K)
- T = Absolute temperature (°K)
- d_{pa} = Particle aerodynamic diameter (micrometers)
- μ = Gas viscosity (kg/m·sec)

**Mechanism of significant
for $D_p < 0.3 \mu\text{m}$
(esp. $0.01 - 0.1 \mu\text{m}$).**

BROWNIAN DIFFUSION – Contd.

Mechanism is characterized by the diffusional deposition number, N_D

where
$$N_D = \frac{D_d}{v \cdot D_c} = \frac{1}{P_e}$$

D_d is diffusivity

v is the gas velocity

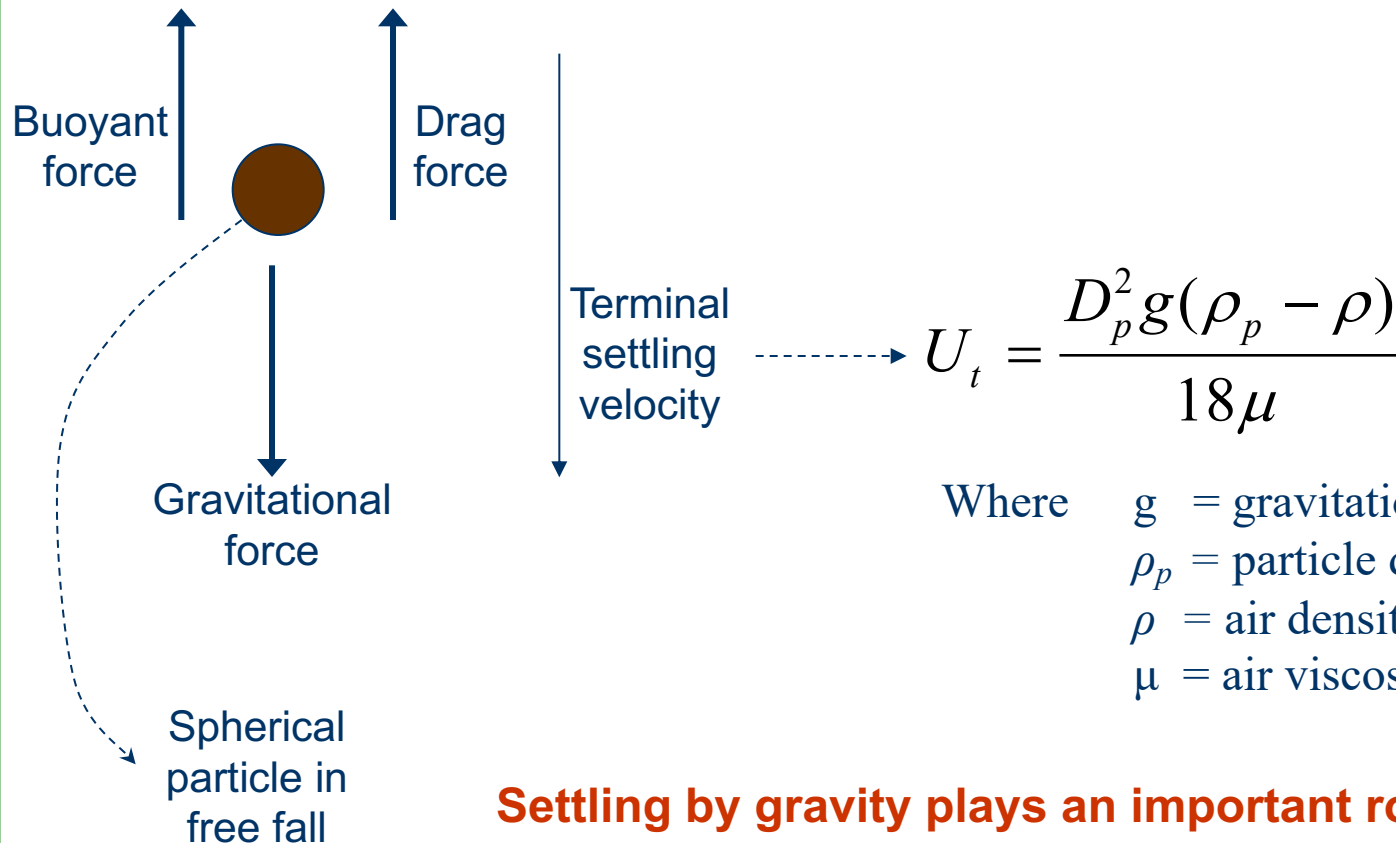
D_c is the diameter of the collector

P_e is the Peclet number

GRAVITATIONAL SETTLING

Particles in still air have three forces acting on them;

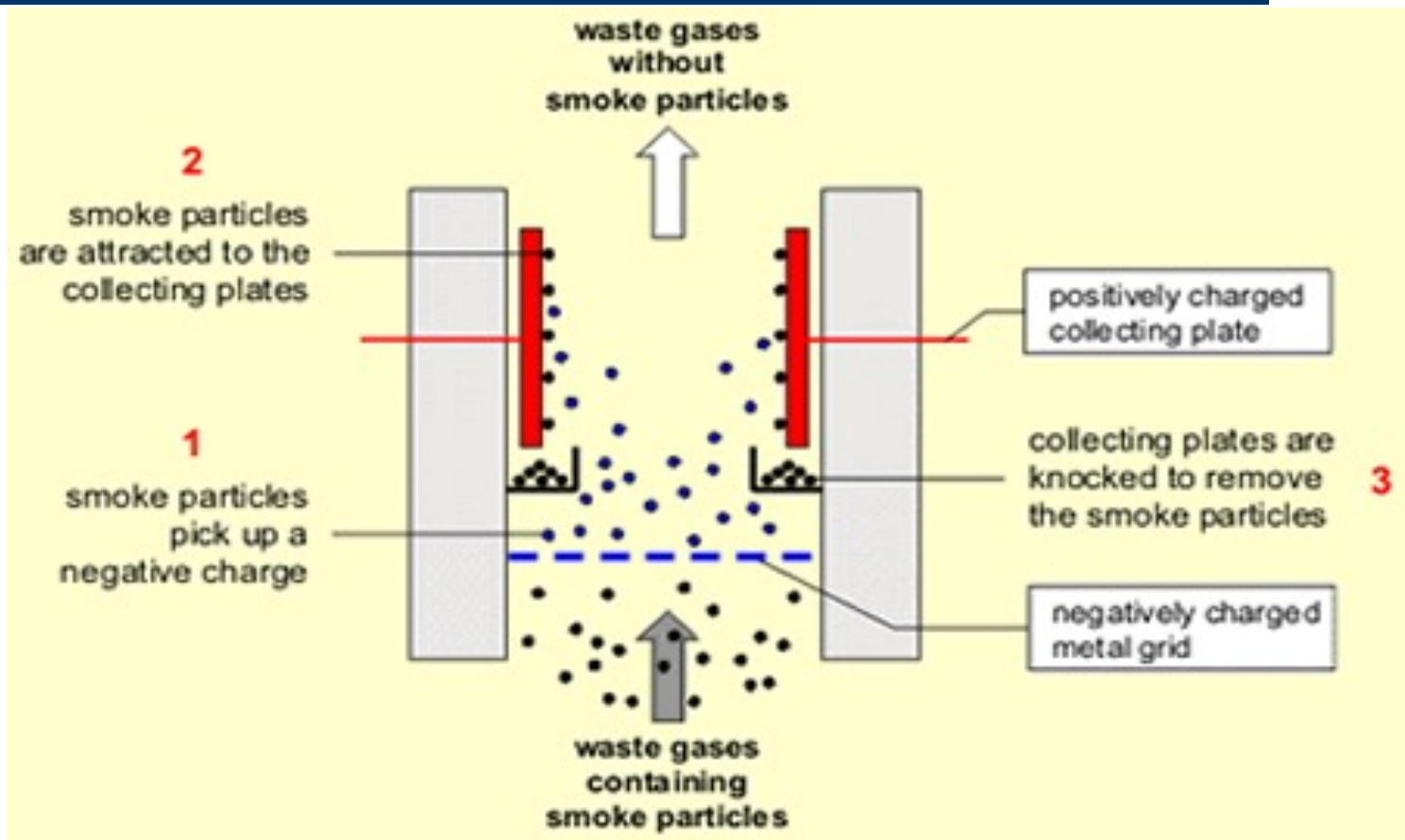
1. **gravitational force downward,**
2. **the air resistance (or drag) force upward and**
3. **buoyant force (by the displaced air) upwards.**



**Settling by gravity plays an important role
when $D_p > 40\mu\text{m}$**

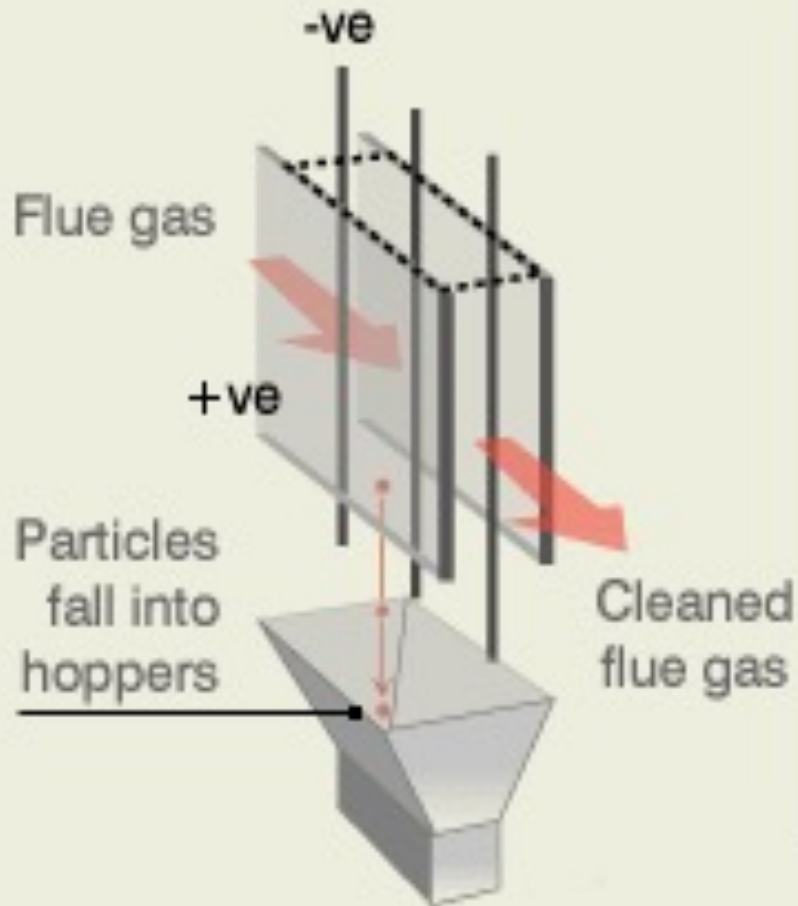
ELECTROSTATIC ATTRACTION

Electrostatic attraction of particles is accomplished by establishing a strong electrical field and creating unlike charges.



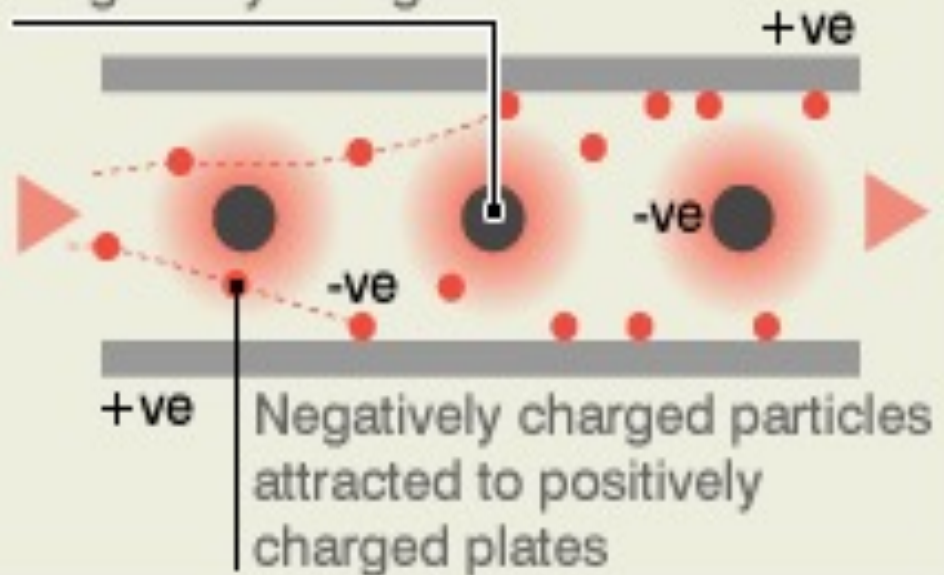
ELECTROSTATIC ATTRACTION – contd.

ELECTROSTATIC PRECIPITATION



CROSS SECTION

Electrodes cause particles to become negatively charged



SOURCE: Powerspan Corp.

ELECTROSTATIC ATTRACTION – contd.

The electrical migration velocity, v_e is given by:

$$v_e = \frac{E_o E_p D_p C_c}{\pi \mu}$$

Where

E_o = charging field strength (volts/m)

E_p = collecting field strength (volts/m)

μ = gas viscosity g/(cm) (sec)

C_c = Cunningham's correction factor

When $E_o = E_p$ as in slide 33, then

$$v_e = \frac{E^2 D_p C_c}{\pi \mu}$$

Thermophoresis and Diffusiophoresis

- **Thermophoresis** is particle movement caused by *thermal* differences
- **Diffusiophoresis** is particle movement caused by *concentration* differences